Exercise 3 - SQL Fundamental (CASE Statements)

Question 1

```
SELECT  product_name,
        price
CASE    WHERE price >1000 THEN 'Expensive
        WHERE price BETWEEN 100 AND 1000 THEN 'Mid-range'
        ELSE 'Budget'
        END AS price_category
FROM    products ;
```

product_name	price	price_category
Laptop	1200.00	Expensive
Phone	800.00	Mid-range
Keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

Question 2

```
SELECT  customer_name,
        amount
CASE WHERE  amount ≥ 1000 THEN 'High Value'
     WHERE amount BETWEEN 500 AND 999.99 THEN 'Medium
     Value'
     ELSE 'Low Value'
     END AS order_value_category
FROM orders ;
```

customer_name	amount	order_value_category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	Medium Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

Question 3

```
SELECT  emp-name,
        department,
        salary
CASE WHERE department='IT' AND salary>80000 THEN 'Senior IT'
     WHERE department='HR' AND salary>55000 THEN 'Experienced HR'
     ELSE 'Staff'
     END AS position_level
FROM  employees ;
```

emp-name	department	salary	position_level
John	IT	85000	Senior IT
Sarah	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

→ WHERE our statement would filter out 'Finance' but CASE doesn't filter.

Question 4

```
SELECT  student_name,
        score
CASE  WHERE score ≥ 90 THEN 'A'
      WHERE score BETWEEN 80 AND 89 THEN 'B'
      WHERE score BETWEEN 70 AND 79 THEN 'C'
```

```
        WHERE score BETWEEN 60 AND 69 THEN 'D'
        ELSE 'F'
        END AS "grade"
FROM students;
```

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	569	F
David	83	B
Ella	68	D

Question 5

```
SELECT delivery_id,
       delivery_time_minutes
CASE WHERE delivery_time_minutes ≤ 30 THEN 'Fast'
     WHERE delivery_time_minutes BETWEEN 31 AND 60 THEN
     'On Time'
     ELSE 'Late
END AS performance
FROM deliveries
```

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late.

Question 6

```
SELECT issue_type,
       priority
CASE WHERE priority = 3 THEN 'High'
     WHERE priority = 2 THEN 'Medium'
     ELSE 'Low'
END AS priority_label
FROM tickets;
```

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password Reset	1	Low

Question 7

```
SELECT student_id,
       (days_present / total_days * 100) AS attendance_percentage
CASE WHERE (days_present/total_days * 100) ≥ 90% THEN 'Excellent'
     WHERE(days_present/total_days * 100) BETWEEN 75 AND 89
     THEN 'Good'
     ELSE 'Needs Improvement'
END AS attendance_status
FROM attendance;
```

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Need Improvement
5	100	Excellent

Question 8

```
SELECT product_id,
       stock_qty
CASE WHERE stock_qty = 0 THEN 'Out-of-Stock'
     WHERE stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
     ELSE 'In Stock'
END AS stock_status
FROM products_inventory;
```

product_id	stock_qty	stock_status
1	5	In Stock
2	0	Out-of-Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

Question 9

```
SELECT subject,
       enrolled_students
CASE WHERE enrolled_students ≥ 25 THEN 'Large'
     WHERE enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
     ELSE 'Small'
END AS class_size_category
FROM classes;
```

subject	enrolled_students	class_size_category
Maths	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

Question 10

```
SELECT ~~payment~~ *
CASE WHERE payment_method = 'Cash' AND amount ≥ 200 THEN
        'Eligible for Discount'
        ELSE 'Not Eligible'
END AS discount_eligibility
FROM payments;
```

payment_id	amount	payment_method	discount_eligibility
1	50.00	Card	Not Eligible
2	200.00	Cash	Eligible for Discount
3	150.00	Card	Not Eligible
4	75.00	PayPal	Not Eligible
5	300.00	Cash	Eligible for Discount